When Self-Correction Fails: Diagnosing Retrieval Coverage in Hallucination Mitigation for Large Language Models

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***Abstract*—We present a self-correcting framework that decomposes a generated answer into atomic claims, retrieves evidence for each claim independently, verifies each claim with a natural language inference model, and revises only the claims that fail, committing a revision only when it raises the aggregate support score. We evaluate the framework against a vanilla LLM and a retrieval-augmented baseline on TruthfulQA, HaluEval and FEVER, and report a mixed and largely negative outcome. As a *detector*, evidence-grounded verification is dramatically better than LLM self-assessment: on HaluEval it attains $F_1 = 0.583$ against 0.066 for the vanilla baseline, whose recall of 3.9% shows that the model almost never identifies a hallucination it is shown. As a *generator*, the same machinery is harmful: it loses 9.8 pp to the vanilla baseline on TruthfulQA ($p_{\text{adj}} = 0.027$) and 11.6 pp on FEVER ($p_{\text{adj}} = 0.022$). Our ablation shows the monotonic acceptance gate, the component we introduced to prevent degradation, itself costs 5.7 pp. Diagnostic analysis identifies retrieval coverage, rather than verification logic, as the principal factor under the tested conditions: 93.8% of TruthfulQA queries retrieve a top passage below 0.5 cosine similarity and gold evidence recall on FEVER is 10.8%, so claims are marked unsupported because evidence is absent, not because they are false. We conclude that, in the regime we tested, claim-level gated self-correction is contingent on retrieval quality in a way the self-correction literature does not adequately report, and we quantify the coverage conditions under which it fails. Single backbone, single seed and a reduced corpus mean this is an identified association rather than an established cause.**

***Index Terms*—large language models, hallucination detection, retrieval-augmented generation, natural language inference, negative results**

I. Introduction

Large language models generate fluent text irrespective of whether the underlying assertions are true, a failure mode termed *hallucination* [1]. Retrieval-augmented generation [2] supplies correct information in context but imposes no constraint that the output be entailed by it. Self-refinement methods [3], [4] ask the model to revise its own output, but Huang et al. [5] show that without external feedback such loops frequently fail to improve and sometimes degrade the answer.

This paper reports an attempt to close that gap by making verification explicit, claim-level and gated, and an evaluation that did not support the approach for generation. Our contributions are:

1) A framework combining atomic claim decomposition, claim-conditioned retrieval, NLI verification, and a *monotonic acceptance gate* that commits a revision only when it improves measured support. We verify the resulting invariant empirically: it held on all 194 generations.
2) Evidence that this machinery is a strong *detector* and a poor *generator*. On HaluEval it raises detection F_1 from 0.066 to 0.583; on TruthfulQA and FEVER it significantly underperforms an unaugmented LLM.
3) A diagnosis. Under the conditions tested, both losses track retrieval coverage rather than the verification logic, and we report the coverage statistics that make this identifiable. We argue these statistics should be standard in self-correction papers, since without them a coverage failure is indistinguishable from a method failure.
4) A negative ablation result: the acceptance gate costs 5.7 pp on the generation benchmark. We report it because it contradicts our own design rationale and because the mechanism, lock-in of hedged drafts, is informative.

II. Related Work

Measuring hallucination. Ji et al. [1] distinguish intrinsic hallucination (contradicting the source) from extrinsic (unverifiable against it); our claim verdicts map onto this distinction. TruthfulQA [6] targets imitative falsehoods and reports truthfulness and informativeness separately, since refusal trivially maximises the former. HaluEval [7] frames detection as balanced binary classification. FEVER [8] supplies three-way labels with gold evidence, allowing retrieval quality to be measured independently of verification quality, which proves essential to our analysis.

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Detection without gold evidence. SelfCheckGPT [9] treats inconsistency across samples as a hallucination signal, detecting instability rather than falsity. FActScore [10] establishes atomic decomposition as the right granularity for factuality assessment; we adopt that granularity as a control signal for correction rather than only as a metric.

Correcting generated text. RARR [11] edits outputs to agree with post-hoc retrieved evidence. Chain-of-Verification [12] plans and answers verification questions. CRITIC [13] critiques via external tools. Self-RAG [14] trains reflection tokens governing retrieval and support. These works report aggregate gains; none, to our knowledge, reports the retrieval coverage under which those gains were obtained. Our results suggest that omission matters.

Irrelevant retrieval. Our finding has direct precedent. Shi et al. [15] show irrelevant context degrades LLM reasoning, and Yoran et al. [16] show retrieval-augmented models are damaged by passages that do not bear on the question, proposing NLI-based filtering as a remedy. Our RAG baseline's -27.8 pp collapse on FEVER is an instance of the same effect, and our UNVERIFIABLE verdict (Section VII) applies that filtering per claim rather than per prompt.

Deciding when to retrieve. Mallen et al. [17] show retrieval helps on rare entities and hurts on common ones, selecting on entity popularity; FLARE [18] triggers retrieval when token confidence drops; Self-RAG [14] learns the decision. These motivate the routing of Section VII, whose signals draw on work showing model confidence carries usable information about correctness [19], [20]. Our retriever follows standard practice [21].

Entailment for verification. We use an MNLI-trained [22] DeBERTa model [23], aggregating entailment and contradiction asymmetrically over retrieved passages.

III. Method

A. Pipeline

Algorithm 1 gives the loop. Given question q: retrieve top-k passages and generate a draft a_0; decompose a_0 into atomic claims $C = \{c_1, \ldots, c_m\}$ by few-shot prompting [10]; retrieve k_c passages E_i per claim; score each claim by NLI; revise flagged claims; re-verify; accept or reject.

B. Claim-Conditioned Retrieval

Evidence is retrieved once per claim rather than once per question. A question-level query returns passages about the topic; a claim-level query returns passages about the specific assertion, which is what an entailment model requires. Maximal Marginal Relevance re-ranking is applied because near-duplicate passages otherwise produce falsely confident support.

C. Verification

The support score aggregates asymmetrically:

$$s(c_i) = \max_{e \in E_i} P(\text{entail} \mid e, c_i) - \max_{e \in E_i} P(\text{contradict} \mid e, c_i). \quad (1)$$

Algorithm 1 Self-correcting generation

$a \leftarrow \textsc{Generate}(q, \textsc{Retrieve}(q, k))$; $a_0 \leftarrow a$
$R \leftarrow \textsc{Verify}(\textsc{Decompose}(q, a), q)$; $S^\star \leftarrow S(R)$
for $t = 1$ **to** T **do**
  if all claims SUPPORTED **then**
    break
  end if
  $a' \leftarrow \textsc{Revise}(q, a, \textsc{Flagged}(R))$
  $R' \leftarrow \textsc{Verify}(\textsc{Decompose}(q, a'), q)$
  if $S(R') \geq S^\star + \varepsilon$ **then**
    $a \leftarrow a'$; $R \leftarrow R'$; $S^\star \leftarrow S(R')$
  else
    break
  end if
end for
return a, R

Maxima rather than means: averaging lets a pool of irrelevant neutral passages dilute a single decisive contradiction. Claims receive a three-way verdict against thresholds τ_s and τ_r, and the answer-level score is $S(a) = \frac{1}{m}\sum_i s(c_i)$.

a) Calibration: τ_s is selected on a held-out development set of 200 HaluEval items disjoint from all test splits, by maximising *balanced accuracy*. We report this choice explicitly because maximising F_1 is unsound here: on a class-balanced detection set the degenerate classifier flagging every claim attains recall 1.0 and precision equal to the base rate, giving $F_1 \approx 0.67$, higher than most discriminative thresholds. In preliminary experiments F_1 calibration selected $\tau_s = +0.89$ with a 96% flag rate and chance-level accuracy. Balanced accuracy selected $\tau_s = -0.71$ with a 46.5% flag rate.

D. Targeted Revision and the Acceptance Gate

Flagged claims and their evidence are returned to the LLM with an instruction to correct or remove them while preserving verified content. A candidate revision a' is committed only if

$$S(a') \geq S(a) + \varepsilon, \quad (2)$$

otherwise the loop terminates. Equation (2) yields the invariant $S(a_{\text{final}}) \geq S(a_0)$: the returned answer never scores below its own draft. Section V-D shows this invariant holds empirically and that it nonetheless harms end-task accuracy.

E. Task Adaptation

For HaluEval the candidate answer is supplied, so the pipeline runs verification only. For FEVER the three NLI verdicts map onto the three FEVER labels and no generation occurs.

IV. Experimental Setup

Data. TruthfulQA [6] generation split (194 test items), HaluEval [7] QA subset (207), FEVER [8] labelled development split (198). Thresholds were calibrated on 200 HaluEval items disjoint from every test split.

Models. Generation: Qwen2.5-7B-Instruct, 4-bit NF4, greedy decoding. Embeddings: `all-mpnet-base-v2` [24]. Verification: `deberta-large-mnli` [23]. Retrieval: FAISS [25] exact inner product, $k = 5$, $k_c = 3$. Hardware: one A100-80GB.

Corpus. The evidence pool comprises FEVER gold-evidence pages plus 20,000 randomly sampled distractor Wikipedia pages, chunked into 26,099 passages of 100 words with overlap. *This is a reduced pool, not full-Wikipedia retrieval.* Its limitations are central to our findings and quantified in Section V-C.

Metrics. TruthfulQA: truthfulness, informativeness and their conjunction. Truthfulness uses reference-similarity scoring, a *proxy* for the canonical GPT-judge metric [6], whose judge models are no longer publicly served; our numbers are therefore not comparable to published GPT-judge scores. HaluEval: detection accuracy, precision, recall, F_1. FEVER: label accuracy, macro-F_1, gold evidence recall.

Reproducibility. The corpus streams the FEVER Wikipedia dump, keeping every gold-evidence page for the sampled claims plus 20,000 randomly encountered distractor pages; pages are stripped of FEVER markup and split by a 100-word sliding window with stride 80, giving 26,099 passages whose identifiers keep the raw wiki title so gold-evidence matching is exact. Embeddings are L2-normalised and indexed with exact inner product. For the TruthfulQA proxy we embed each answer a with the item's reference sets and compute $s_+ = \max_{r \in R_+} \cos(a, r)$, $s_- = \max_{r \in R_-} \cos(a, r)$; a is truthful when $s_+ > s_-$ and $s_+ \geq 0.60$, and informative unless it contains one of eight fixed refusal markers ("i have no comment", "i don't know", "i do not know", "i'm not sure", "no comment", "cannot be determined", "evidence is insufficient", "i cannot answer"). Neither the floor nor the marker list was tuned. Decoding is greedy, so generation is deterministic given the prompt; we did not vary the seed governing sampling and the dev/test split, so intervals reflect variability within one split, not across splits. Environment: one NVIDIA A100-80GB, PyTorch 2.11, Transformers 5.16.

Statistics. All comparisons are paired on identical items. Binary per-item metrics use the exact McNemar test; Holm–Bonferroni correction is applied across the comparison family; 95% percentile bootstrap confidence intervals use 10,000 resamples. Single seed (42); see Section VIII.

V. Results

A. Detection versus Generation

Table I shows the framework is significantly worse than an unaugmented LLM on both generation-oriented benchmarks and better, though not significantly so after correction, on detection.

The detection result is more informative than its accuracy suggests (Table II). Both baselines are effectively inert: the vanilla LLM identifies 4 of 102 hallucinated answers (recall 3.9%) and RAG identifies 2 (recall 2.0%). Asked whether text it is shown contains a false claim, the model almost always answers no. Evidence-grounded verification reaches recall 61.8% at precision 55.3%, raising F_1 from 0.066 to 0.583, close to an order of magnitude. The modest accuracy gain understates this: accuracy on a balanced set rewards the baselines' correct rejection of the 105 clean answers.

TABLE I
Main results. Percentages with 95% bootstrap CIs. Significance against the vanilla baseline, exact McNemar with Holm–Bonferroni correction.

Benchmark	System	Primary metric	Δ vs. vanilla
TruthfulQA	Vanilla	51.0 [44.3, 58.2]	–
	RAG	27.8 [21.6, 34.0]	-23.2^{***}
	Proposed	41.2 [34.0, 48.5]	-9.8^{*}
HaluEval	Vanilla	45.4 [38.6, 52.2]	–
	RAG	49.8 [43.0, 56.5]	$+4.3$
	Proposed	**56.5** [49.8, 63.3]	$+11.1$
FEVER	Vanilla	**63.1** [56.1, 69.7]	–
	RAG	35.4 [28.8, 41.9]	-27.8^{***}
	Proposed	51.5 [44.4, 58.6]	-11.6^{*}

$^{*}p_{\mathrm{adj}} < 0.05$, $^{***}p_{\mathrm{adj}} < 0.001$; HaluEval gains n.s. ($p_{\mathrm{adj}} = 0.069$)

TABLE II
HaluEval detection. The baselines almost never flag anything.

System	Acc.	Prec.	Rec.	F_1	TP/FP/FN/TN
Vanilla	45.4	21.1	3.9	0.066	4/15/98/90
RAG	49.8	33.3	2.0	0.037	2/4/100/101
Proposed	**56.5**	55.3	**61.8**	**0.583**	63/51/39/54

TABLE III
TruthfulQA decomposition (%). The loss is entirely informativeness.

System	Truthful	Informative	Conjunction	Refusal rate
Vanilla	51.0	100.0	51.0	0.4
RAG	49.5	58.8	27.8	77.1
Proposed	49.5	83.5	41.2	62.9

B. Where the Generation Loss Comes From

Decomposing the TruthfulQA metric (Table III) localises the failure. Truthfulness is statistically indistinguishable across all three systems (51.0, 49.5, 49.5). The entire difference is informativeness: the vanilla baseline refuses on 0.4% of questions, RAG on 77.1%, and the proposed system on 62.9%. Because the primary metric is the conjunction (a model can trivially maximise truthfulness by declining to answer), refusal drives the loss.

C. Retrieval Coverage as the Principal Identified Factor

Retrieval diagnostics (Fig. 1) are consistent with the refusals being driven by coverage. On TruthfulQA, 93.8% of queries retrieve a top-ranked passage below 0.5 cosine similarity (median 0.360, 90th percentile 0.456). TruthfulQA targets misconceptions about the brain, folk superstitions and common myths, for which a 26,099-passage Wikipedia subset holds little relevant material. Every claim is then scored unsupported,

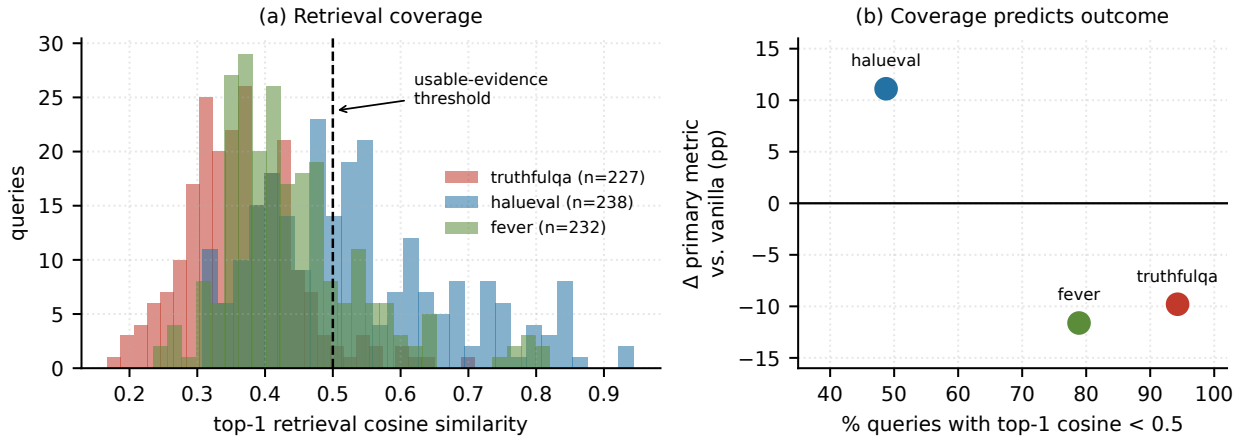


Fig. 1. (a) Distribution of top-1 retrieval similarity per benchmark. The TruthfulQA mass lies almost entirely below the usable-evidence threshold. (b) Per-benchmark change in the primary metric against the fraction of weakly retrieved queries. The ordering is consistent with retrieval coverage driving the outcome, but three benchmarks are three points: this is a visual summary of Sections V-B–V-C, not a fitted relationship, and no trend should be read into it.

the revision prompt drives the answer toward explicit uncertainty, and informativeness collapses.

On FEVER, gold evidence recall is 10.8% for RAG and 10.1% for the proposed system, with 86.8% of queries below 0.5 similarity. A claim cannot be verified against evidence that was never retrieved, so this figure upper-bounds achievable verification accuracy. The RAG baseline's −27.8 pp collapse on FEVER has the same cause: conditioning a model on irrelevant passages is worse than not conditioning it at all.

D. Acceptance Gate

The gate behaved exactly as designed and still hurt. Of 146 candidate revisions, 50 were committed (34.2%) and 96 rejected. Accepted revisions raised aggregate support by +0.163 on average; rejected candidates would have changed it by −0.115. The monotonicity invariant $S(a_{\text{final}}) \geq S(a_0)$ held on all 194 generations.

Yet the ablation (Table IV) shows removing the gate *improves* the primary metric by 5.7 pp. The mechanism is visible in the qualitative categories: 118 of 194 cases are refusals in both draft and final answer, and 26 are cases where the gate blocked a revision. Under low retrieval coverage the draft is already a hedge; the gate, which admits only revisions that raise measured support, locks that hedge in. Support and informativeness are anti-correlated in this regime, so optimising the former degrades the latter.

Claim decomposition and iteration contribute 1.5 pp each and claim-conditioned retrieval 0.5 pp, unsurprising when the corpus lacks relevant passages.

a) On the two pairs of equal values: The metric is a per-example binary outcome over 194 items, so arm means are counts $k/194$ and ties are unremarkable across seven arms. The two pairs arise differently. Dropping claim-conditioned retrieval and restricting the loop to one iteration are independent runs whose answers differ on 54 of 194 items and whose revision activity differs (48/139 versus 48/123 accepted); they coincide at 79 successes by chance. Dropping decomposition and disabling revision differ on one answer only, for a mechanical reason: without decomposition the whole answer is a single claim, the gate then accepts 1 of 5 candidate revisions, and the arm collapses onto no-revision behaviour. Atomic decomposition is what lets the loop act at all.

TABLE IV
ABLATION ON TRUTHFULQA. POSITIVE Δ MEANS REMOVING THE COMPONENT HELPS. THE PRIMARY METRIC IS A PER-EXAMPLE BINARY OUTCOME, SO WE ALSO GIVE THE RAW COUNT OF SUCCESSES OUT OF 194; THE TWO PAIRS OF EQUAL PERCENTAGES ARE EXACT TIES IN THAT COUNT, NOT DUPLICATED RUNS (SEE TEXT).

Variant	Primary	Count	Δ	Rev. acc./att.
Full framework	41.2	80/194	–	50/146
w/o acceptance gate	**46.9**	91/194	+5.7	298/298
revise all claims	43.3	84/194	+2.1	69/157
w/o claim retrieval	40.7	79/194	−0.5	48/139
single iteration	40.7	79/194	−0.5	48/123
w/o decomposition	39.7	77/194	−1.5	1/5
verification only	39.7	77/194	−1.5	0/0

E. Cost

The framework issues 3.51 LLM calls, 7.20 retrieval calls and 18.6 NLI calls per TruthfulQA query, at 9.13 s mean latency, against one LLM call and 3.00 s for the vanilla baseline. On this evidence the accuracy-per-unit compute is negative for generation.

VI. Discussion

A. Qualitative Categories

Fig. 2 categorises every generation by draft/final pair. Refusals in both dominate at 118 of 194; a further 26 are gate-blocked revisions, only 8 were revised and kept, and 4 were revised from an informative answer into a refusal. The

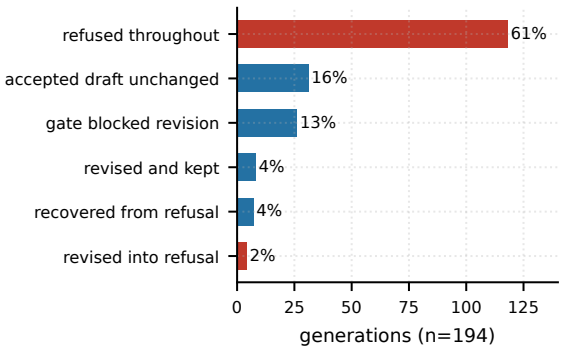


Fig. 2. Deterministic categorisation of all 194 TruthfulQA generations. Assignment is by rule from the draft/final answer pair; no examples were selected by hand. Refusal-related categories (dark) account for 63% of cases, which is the qualitative signature of the coverage failure quantified in Fig. 1.

framework spends its compute overwhelmingly on answers it cannot ground rather than ones it can repair.

B. An Implementation Fault We Report Rather Than Hide

Our calibration procedure selected $\tau_r = \tau_s = -0.71$, collapsing the three-way verdict to two-way and making NOT ENOUGH INFO unreachable. The FEVER confusion matrix confirms this: no item received an NEI prediction, so all 50 NEI gold items were necessarily wrong. FEVER accuracy of 51.5% is therefore an underestimate of the method's ceiling, and the fix, constraining τ_r to lie strictly below τ_s, is straightforward. We report the affected number rather than a re-run one because we did not re-run it; readers should treat the FEVER column as a lower bound.

C. Implications

Three claims follow from these results, each scoped to the single-backbone, single-seed, reduced-corpus setting we tested.

First, *evidence-grounded verification is a genuinely better detector than LLM self-assessment*. The gap between $F_1 = 0.066$ and 0.583 is not marginal and holds under a poor retrieval regime. Methods relying on a model to judge its own output should report recall, not only accuracy; at recall 3.9% the vanilla baseline is not performing the task at all.

Second, *self-correction for generation appears contingent on retrieval coverage in a way the literature under-reports*. We could not find coverage statistics such as retrieval similarity distributions or gold evidence recall in the self-correction papers we surveyed; without them a coverage failure is indistinguishable from a method failure. We recommend both be reported as standard.

Third, *a gate that enforces monotonicity in a proxy metric can degrade the true objective*. Our gate provably never lowers measured support and, in our runs, lowers end-task accuracy. This is not a contradiction but the gap between support and factuality made measurable. Guarantees stated over proxy metrics should come with evidence that the proxy tracks the objective in the operating regime.

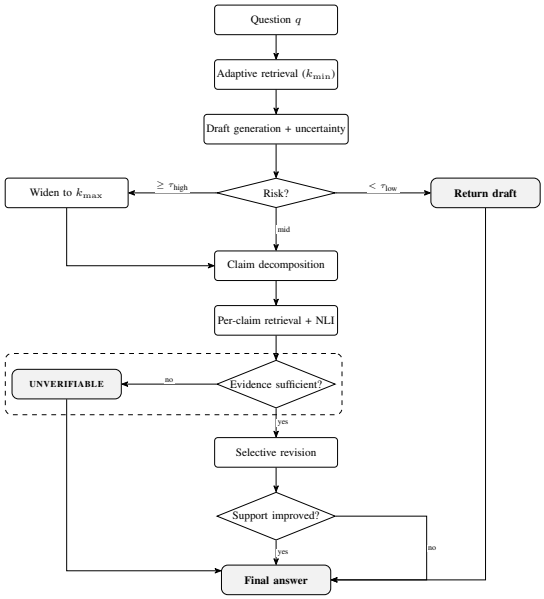


Fig. 3. Adaptive Self-Correcting RAG. Risk estimation routes each query to one of three paths, so low-risk answers incur no verification cost. The dashed guard separates *no evidence retrieved* from *claim is false*: claims below the retrieval floor are marked UNVERIFIABLE and left unmodified rather than revised into hedges.

VII. Proposed Extension: Adaptive Correction

Our diagnosis identifies a specific defect: the pipeline assigns the same verdict, and takes the same corrective action, when a claim is false and when no evidence for it was retrieved. Under 10.8% evidence recall this conflation is the dominant failure mode. We describe the extension it motivates. **It is a design, not a result: the system described here is implemented but not yet evaluated, and no number in this paper comes from it.**

Fig. 3 shows the architecture. Three changes follow directly from the failure modes in Section VIII.

A retrieval-failure guard. A fourth verdict, UNVERIFIABLE, is assigned when the retrieved evidence for a claim falls below a confidence floor. Unverifiable claims are excluded from revision and from the aggregate support score, so an answer is neither rewritten nor penalised because the retriever failed. This targets the 118 of 194 cases we categorised as refusals in both draft and final answer.

Risk-conditioned routing. A logistic model over eight cheap signals (NLI entailment and contradiction, answer–evidence similarity, retrieval confidence and spread, token log-probability, predictive entropy and lexical hedging) estimates hallucination risk before verification is attempted. Low-risk answers are returned unmodified at the cost of one LLM call, which addresses the negative accuracy-per-compute reported in Section V-F. Only high-risk answers pay for widened retrieval and correction.

Adaptive retrieval depth. Top-k expands from 3 to 10 when the top-ranked passage is weak, and the query is abandoned as unverifiable when widening does not help.

Whether these changes recover the generation loss is untested. The prediction is specific: if the loss is caused by conflating absent evidence with false claims, the guard should raise informativeness toward the vanilla baseline's 100% without lowering truthfulness below the 49.5% observed here. If informativeness does not move, our diagnosis is wrong.

VIII. Limitations

Single seed, single backbone (Qwen2.5-7B-Instruct) and a reduced retrieval pool each limit generality; multi-seed and multi-backbone replication is the immediate next step, and until it is done our coverage account is an association identified under these conditions rather than an established cause. The TruthfulQA metric is a similarity proxy, not the canonical judge, and the τ_r collapse makes FEVER a lower bound. Our negative finding is specific to the low-coverage regime measured and does not establish that gated self-correction fails under strong retrieval. Sentence-level NLI is weak on multi-hop and numeric claims. HaluEval gains did not survive Holm–Bonferroni correction ($p_{\mathrm{adj}} = 0.069$) and are suggestive only. The extension of Section VII is unevaluated.

IX. Conclusion

We evaluated a claim-level, evidence-grounded, gated self-correction framework and found it improves hallucination *detection* substantially (F_1 $0.066 \rightarrow 0.583$) while significantly harming factual *generation* (-9.8 pp on TruthfulQA, -11.6 pp on FEVER). Diagnostic analysis points to retrieval coverage (93.8% of TruthfulQA queries retrieve nothing relevant, and FEVER gold evidence recall is 10.8%), rather than verification logic, as the principal factor we could identify, and our ablation shows the acceptance gate we introduced costs 5.7 pp by locking in hedged drafts. Under the conditions we tested, the evidence pool rather than the entailment model appears to be the limiting factor. We suggest coverage statistics become standard in this literature. Section VII describes the adaptive architecture this diagnosis motivates, whose evaluation is our immediate next step.

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